

TSBC-1502 - Stage 12 blocked-dedup recovery churn fix

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Source issue: `/TSBC/issues/TSBC-1502`

PDF filename: `TSBC-1502-report.pdf`

Hypothesis verdict: `CONFIRMED`

Dispatched follow-up key: `TSBC-1404`

Catalog row: `tsbc-1502-stage12-blocked-dedup-noop`

Report date: `2026-07-30`

Brand pack: `stack-lab`

Adoption status: `benchmark_only`

Readiness gate: `report_ready`

Claims gate: `approved_internal`

Hypothesis

A deliberate Stage 12 blocked-dedup / no-op summary should classify as `blocked`, preserve the blocked disposition without comment churn, skip successful-run-handoff recovery, and avoid stale reassignment loops.

Method

Validation used the served Paperclip tree at `/Users/glad0s/paperclip`, where the live server on port `3100` is running.

Evidence steps:

1. Verified the served liveness/routing implementation already contains the blocked-dedup classifier and blocked-disposition preservation path in `server/src/services/run-liveness.ts`, `server/src/services/blocked-dedup-noop.ts`, and `server/src/services/heartbeat.ts`.
2. Added a focused served-tree regression in `server/src/__tests__/heartbeat-process-recovery.test.ts` that seeds the exact Stage 12 fixture wording, the prior blocked-status comment, and an assertion that the adapter is not re-invoked after the no-op summary.

3. Re-ran the tight served-tree verification set:

- `server/src/__tests__/heartbeat-auto-checkout.test.ts`
- `server/src/__tests__/heartbeat-stale-queue-invalidation.test.ts`
- `server/src/__tests__/heartbeat-process-recovery.test.ts -t "stage 12 blocked-dedup no-op"`

4. Attempted a bounded live rerun from the frozen `[TSBC-1262] (/TSBC/issues/TSBC-1262)` wrapper with:

```
TSBC_R1_REPS=1 TSBC_R1_TASK_LIMIT=12 TSBC_R1_ALLOW_DUPLICATE_STAGE12=1 \  
TSBC_R1_REPORT_STEM=TSBC-1502-stage12-rerun-2026-07-30 \  
python3 -u run-paperclip-r1.py
```

Per the canonical TSKB delta cited below, `TSBC_R1_TASK_LIMIT` is prefix-only, so this live attempt is supplemental evidence rather than a true Stage-12-only selector.

Data

Primary confirming evidence:

- Historical failing Stage 12 chain:
 - source run `734f8a83-f229-4c56-b5ef-b83b6aa2f35c`
 - corrective run `f5301752-f597-44ac-a04f-8a8dfca5cbb9`
 - follow-on assignment run `93b52803-1483-408d-a32b-f5419dd5dcfc`
- Served-tree code paths:
 - `server/src/services/run-liveness.ts`
 - `server/src/services/blocked-dedup-noop.ts`
 - `server/src/services/heartbeat.ts`
- Served-tree targeted regression/logs:
 - `work-products/TSBC-1502/evidence/vitest-heartbeat-routing.log`
 - `work-products/TSBC-1502/evidence/vitest-stage12-blocked-dedup.log`

Observed served verification results:

- `vitest-heartbeat-routing.log`: Test Files 2 passed, Tests 31 passed
- `vitest-stage12-blocked-dedup.log`: Test Files 1 passed, Tests 1 passed
- The focused Stage 12 regression proves:
 - no successful-run-handoff wake is queued;
 - no recovery notice comment is created;
 - the issue remains `blocked`;

- the adapter executes only once for the source run.

Supplemental bounded live rerun evidence:

- Rerun root: `/Users/glad0s/.paperclip/instances/default/projects/e212ce50-b524-408c-b3d4-0c6108d8c2e2/f71e8665-3f38-4920-b777-348ec85b9071/_default/work-products/TSBC-1141/TSBC-1262/runs/paperclip-20260730-052221`
- Sample namespace: `/Users/glad0s/.paperclip/instances/default/projects/e212ce50-b524-408c-b3d4-0c6108d8c2e2/f71e8665-3f38-4920-b777-348ec85b9071/_default/work-products/TSBC-1141/sample-runs/TSBC-1502-stage12-rerun-2026-07-30-20260730T052221Z`
- The attempt advanced through sample 9, then was stopped after the earlier Stage 6/Stage 9 fixtures consumed the bounded window before Stage 12:
 - Stage 6 child issue `[TSBC-1508] (/TSBC/issues/TSBC-1508)` resolved a successful-run handoff from source run `186a0e2d-285d-4bba-a428-166660bc53fa` via corrective run `4d3110a5-f5bf-4bbb-b316-28fcb59b5851`.
 - Stage 9 fixture `[TSBC-1509] (/TSBC/issues/TSBC-1509)` was already queued with run `ed575378-93bb-4bb1-8b93-276488c8b57f` when the attempt was stopped.

Hypothesis Verdict

CONFIRMED. The served regression and routing tests prove the targeted Stage 12 path now classifies as **blocked** and stays quiet instead of falling into missing-disposition recovery churn. The bounded live rerun did not reach Stage 12, so it is supplemental and **INCONCLUSIVE** for later-stage runtime proof, but it does not contradict the served fix.

Executive Recommendation

Treat the Stage 12 platform fix as landed and unblock `[TSBC-1404] (/TSBC/issues/TSBC-1404)` for manager-owned closeout work. If a fresh end-to-end proof is still desired, rerun the frozen `TSBC-1262` wrapper only after earlier-stage fixture churn is clear; do not claim that `TSBC_R1_TASK_LIMIT=12` isolates Stage 12 by itself.

Decision class: `benchmark_only`

What Was Tested

Field	Value
OpCo	TSBC / Paperclip platform
Task / role	Stage 12 blocked-dedup no-op liveness routing
Suite	served heartbeat tests + frozen TSBC-1262 bounded rerun attempt
Models	codex_local for served tests; hermes_local grok-4.5 for bounded rerun
Skill pack	paperclip + paperclip-dev
Prompt / agent file	none; served repo implementation and test fixture
Repetitions	1 focused regression; 1 bounded rerun attempt
Judge	issue-state assertions plus preserved run evidence
Scorer calibration	pass
Reproducibility fingerprint	tsbc-1502-stage12-blocked-dedup-noop-2026-07-30

Result Summary

Lane	Score	Score band	Token metric	Adoption read
served-tree targeted regression	1.00	pass	n/a	ready
bounded live rerun	0.50	inconclusive	not_measured	supplemental_only

Benchmark Fairness Closeout

Field	Record
Fairness verdict	pass_with_caveat
Evidence depth	candidate
Run IDs	734f8a83-f229-4c56-b5ef-b83b6aa2f35c , f5301752-f597-44ac-a04f-8a8dfca5cbb9 , 93b52803-1483-408d-a32b-f5419dd5dcfc , 186a0e2d-285d-4bba-a428-166660bc53fa , 4d3110a5-f5bf-4bbb-b316-28fcb59b5851 , ed575378-93bb-4bb1-8b93-276488c8b57f
Repetitions per compared cell	1 targeted rerun attempt; served regression tests
Low-tail / min-score note	not applicable; this is a state-machine regression proof, not a scored content benchmark
Token / cost / runtime note	quality_only; targeted Vitest proof, live rerun runtime only
Scorer caveat	The bounded live rerun did not reach Stage 12 because earlier preregistered stages consumed the prefix-limited task window.
Failure-library IDs	none
Next gate	rerun

Token and Cost Read

quality_only . The acceptance criterion is correctness of liveness routing, not token efficiency. The economically relevant point is that the served tests now prove the no-op path without forcing repeated recovery churn or extra assignment runs.

Failure Analysis

The original defect was a classification gap: Stage 12 emitted useful output describing a deliberate blocked-dedup no-op, but run liveness still labelled the run needs_followup , which triggered successful-run-handoff recovery and a follow-on assignment loop.

The bounded live rerun exposed a separate earlier-stage drag:

- Stage 6 child [TSBC-1508] (/TSBC/issues/TSBC-1508) needed a corrective run before resolving its own successful-run handoff.
- Stage 9 fixture [TSBC-1509] (/TSBC/issues/TSBC-1509) was already queued when the bounded attempt was stopped.

That earlier-stage runtime drag is not evidence against the Stage 12 fix, but it does make the bounded live rerun `INCONCLUSIVE` for later-stage acceptance on its own.

Rollout Guidance

Adoption owner: `Bench-Manager` on `[TSBC-1404](/TSBC/issues/TSBC-1404)`

Source OpCo rollout issue: `[TSBC-1404](/TSBC/issues/TSBC-1404)`

Skill-pack install/attach notes: `none`

Rollback owner and trigger: `Bench-Manager` if future `run-liveness` or `heartbeat` edits reintroduce `successful-run-handoff` churn on blocked `no-op` fixtures

Previous lane / fallback route: `historical failing Stage 12 fixture chain` cited above

Retest trigger: `any future edit to run-liveness, blocked-dedup parsing, heartbeat disposition preservation, or successful-run-handoff routing`

Caveats for derivatives: `internal engineering proof only; do not present the bounded rerun as a Stage-12-only selector or a full-suite closeout`

Privacy And Public-Claim Gate

Control	Value
Data class	<code>internal</code>
Redaction status	<code>not_needed</code>
Allowed evidence use	<code>internal/report</code>
Public claim level	<code>internal_only</code>
Public approval issue	<code>none</code>
Allowed public wording	<code>none</code>

Evidence

Evidence	Link / path
Run id(s)	734f8a83-f229-4c56-b5ef-b83b6aa2f35c , f5301752-f597-44ac-a04f-8a8dfca5cbb9 , 93b52803-1483-408d-a32b-f5419dd5dcfc , 186a0e2d-285d-4bba-a428-166660bc53fa , 4d3110a5-f5bf-4bbb-b316-28fcb59b5851 , ed575378-93bb-4bb1-8b93-276488c8b57f
Source issue(s)	/TSBC/issues/TSBC-1502 , /TSBC/issues/TSBC-1404 , /TSBC/issues/TSBC-1496 , /TSBC/issues/TSBC-1508 , /TSBC/issues/TSBC-1509
Records	work-products/TSBC-1502/evidence/ , .../TSBC-1262/runs/paperclip-20260730-052221/ , .../sample-runs/TSBC-1502-stage12-rerun-2026-07-30-20260730T052221Z/
Skill pack	paperclip , paperclip-dev
Proof bundle manifest	work-products/TSBC-1502/TSBC-1502-report.md
ROI / ledger issue	not_measured
Canonical TSKB delta	/Users/glad0s/TSKB/KB/TSKB0047 [TSBC] - Lane Placement and Skill-Pack Adoption Tracker - v0.1 - 07-06/benchmark-fairness-protocol.md

Render Command

```
~/scripts/brand-suite/brandsuite pdf --brand stack-lab -- --input /Users/glad0s/pa
```